4 The free energy – centre of the thermal universe

Mostly we describe physical systems at fixed temperature and volume (but chemists often work at fixed pressure). The thermodynamic potential which is minimised under those circumstances is the Helmholtz free energy,

$$F(T, V, N) = U - TS \ ,$$

where N is the number of particles (if a gas), U is the internal energy and S is the entropy. F and its partial derivatives allow the **calculation of all thermodynamic functions**: S, pressure p, U, specific heats, C_{v}, etc etc.

How do we calculate it using statistical mechanics? We will address that and illustrate the use of F in the context of the paramagnet and ideal gas.

But firstly let us revise the material from Y2 Thermal Physics on free energies.

4.1 Free energy from thermodynamics

The fundamental definition of the Helmholtz free energy, $F(T, V, N)$, is:

$$F = U - TS \ ,$$

where the "natural variables" for the other functions involved are: $U(S, V, N)$ and $S(U, V, N)$.

The importance of natural variables is illustrated by writing the first law of thermodynamics, for reversible processes, in the form

$$\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V \tag{4.1}$$

and by defintion, $$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_V \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_S \mathrm{d}V \tag{4.2}$$

(Comparing Eq. (4.1) and Eq. (4.2)) $$T = \left(\frac{\partial U}{\partial S}\right)_V \quad \text{and} \quad p = -\left(\frac{\partial U}{\partial V}\right)_S \ ,$$

i.e. we can deduce p and T from $U(S, V, N)$. And from them anything else that might be desired.

For the Helmholtz free energy, $F(T,V,N) = U - TS$, we see (using $\mathrm{d}(fg) = (\mathrm{d}f)g + f\mathrm{d}g$):

$$\begin{aligned}
\mathrm{d}F &= \mathrm{d}U - \mathrm{d}(TS) \\
&= T\,\mathrm{d}S - p\,\mathrm{d}V - S\,\mathrm{d}T - T\,\mathrm{d}S \\
&= -p\,\mathrm{d}V - S\,\mathrm{d}T \\
\Rightarrow \quad & \begin{cases} p = -\left(\dfrac{\partial F}{\partial V}\right)_T \\ S = -\left(\dfrac{\partial F}{\partial T}\right)_V \end{cases}
\end{aligned} \tag{4.3}$$

Important note. *Both* of the expressions in Eq. (4.3) are termed "equations of state". The relation for p is "the" equation of state ($pV = n\mathrm{R}T$ for an ideal gas) – note the left hand side is a function of V and T. The expression for S is sometimes called the *caloric equation of state*. It contains, as a special case, the statement that the internal energy of an *ideal* gas is $U(T,N) = Nu(T)$, where the specific internal energy, $u(T)$, only depends on temperature (as was shown by Joule's experiment). Thus one needs both parts of Eq. (4.3) to specify a thermodynamic system completely.

To obtain the internal energy, U, from F, we may either say directly $U = F + TS$, or in a more indirect manner note that we can write U more in the style of Eq. (4.3), with $\beta = 1/(k_\mathrm{B}T)$:

$$\begin{aligned}
\frac{\partial}{\partial\beta}(\beta F) &= F + \beta\frac{\partial F}{\partial\beta} \\
&= F + \beta\frac{\partial T}{\partial\beta}\underbrace{\frac{\partial F}{\partial T}}_{=-S} \\
&= F + \frac{1}{k_\mathrm{B}T}\frac{\partial}{\partial\beta}\left(\frac{1}{k_\mathrm{B}\beta}\right)(-S) \\
&= F + \frac{1}{k_\mathrm{B}T}(-1)\frac{1}{k_\mathrm{B}\beta^2}(-S) \\
&= F + \frac{1}{k_\mathrm{B}T}(-1)k_\mathrm{B}T^2(-S) \\
&= F + TS \\
&= U\ .
\end{aligned}$$

This is sometimes called the *Gibbs-Helmholtz equation*.

And so on.

4.2 Important aspects of thermodynamic variables

There is an important division of the thermodynamic variables into two types.

Extensive variables which are proportional to the size of the system: entropy (S), volume (V), particle number (N) and energy (U, F etc). The energy group are a little different.
Intensive variables which do not depend on system size: temperature, (T), pressure, (p), and *chemical potential*, (μ), which will be discussed presently.

They occur in *conjugate pairs*: S and T; V and p; and N and μ.

The different energies represent different choices for each pair: $U(S,V,N)$, $H(S,p,N)$ the enthalpy, $F(T,V,N)$ and $G(T,p,N)$, the Gibbs free energy, with the relations

$$\begin{aligned} &U(S,V,N) && H(S,p,N) = U + pV \\ &F(T,V,N) = U(S,V,N) - TS && G(T,p,N) = U - TS + pV \ , \end{aligned}$$

and the differential relations:

$$\begin{aligned} &\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V && \mathrm{d}H = T\,\mathrm{d}S + V\,\mathrm{d}p \\ &\mathrm{d}F = -S\,\mathrm{d}T - p\,\mathrm{d}V && \mathrm{d}G = -S\,\mathrm{d}T + V\,\mathrm{d}p \ . \end{aligned}$$

The operation of going from, for example, U to $F = U - TS$ is a *Legendre transformation* from S and V to T and V. A discussion of Legendre transformations is contained in *Making sense of the Legendre transform*, by Zia, R. K. P. et al, *American journal of physics*, **77**, 614, (2009).

A curious aspect is the following. In thermodynamics one may start from *either* $U(S,V,N)$ or $S(U,V,N)$ (i.e. an inverse function) and deduce all the other quantities. **But** in statistical mechanics one starts from $S(U,V,N)$: I am aware of no starting point from $U(S,V,N)$. This is because (e.g. for a gas) there is a direct relation of dynamics of the constituents and the energy, or $\mathcal{H}$, and then via the ergodic hypothesis to averages over phase space; but there is no counterpart in terms of the entropy of the constituents and a "phase space" for those putative variables, with a corresponding "ergodic hypothesis".

We will encounter the cases involving μ later.

4.3 The free energy and the partition function

What is F in terms of the partition function, $Z(T,V,N)$? The answer is relatively simple:

$$F = -k_{\mathrm{B}} T \ln Z(T,V,N) \ , \text{ compare with } S = k_{\mathrm{B}} \ln W(E,V,N) \ .$$

This emerges from an analysis of the integral for Z, where the integrand has a very sharp maximum as $V \to \infty$ or equivalently as $N \to \infty$. The reason for this is that if the system has N particles (or spins), there are increasingly many ways to partition up E into energy per particle (remember the combinatorial factor in the spins in a magnetic field). Thus $W(E)$

increases very rapidly with E, typically $W(E) \sim E^{\alpha N}$ where α is a constant. Remembering the result in the last subsection, as $N \to \infty$ expect maximum to be very sharp.

Let us rewrite Z as (dropping the subscripts to denote the system – there is no ambiguity here). We retain a subscript on the temperature to remember T_b is a parameter determined by the heat bath.

$$\begin{aligned} Z(T,V,N) &= \int \mathrm{d}E\ W(E) \exp\left(-\frac{E}{k_\mathrm{B}T_\mathsf{b}}\right) \\ &= \int \mathrm{d}E\ \exp\left(\frac{S}{k_\mathrm{B}} - \frac{E}{k_\mathrm{B}T_\mathsf{b}}\right) \\ \left(\text{Defining } \beta_\mathsf{b} = \frac{1}{k_\mathrm{B}T_\mathsf{b}}.\right) \quad &= \int \mathrm{d}E\ \exp\left(-\beta_\mathsf{b}[E - TS(E,V,N)]\right) \ . \end{aligned}$$

where the appearance of $E - TS = F$ in the exponent is very suggestive. Notice that the integral over E has at least one maximum of the integrand.This is because $W(E)$ increases with E and the exponential, $\mathrm{e}^{-\beta_\mathsf{b}E}$, decreases with E. We also suspect this maximum is **very sharp**, because the number of entities involved, N, is very large, and will demonstrate that presently. . Let us find the stationary point, noting the following result.

$$\frac{\mathrm{d}}{\mathrm{d}x} \exp\left[f(x)\right] = f'(x) \exp\left[f(x)\right] = 0 \ ,$$

so as long as $f(x) \neq -\infty$, the maximum is determined by $f'(x) = 0$.

Applying this we see that

$$\begin{aligned} \frac{\partial}{\partial E}\left[E - T_\mathsf{b}S(E)\right] = 1 - T_\mathsf{b}\left(\frac{\partial S}{\partial E}\right)_V &= 0 \\ \Rightarrow \quad \frac{1}{T_\mathsf{b}} &= \frac{1}{T(E,V,N)} \ . \end{aligned}$$

This is an equation for the value of E at the stationary point. I.e. E is determined by the temperature of the *system*, $T(E,V,N)$, being equal to T_b. Let us denote that value of $E = U$.

Then the partition function is approximately, Taylor expanding around $E = U$ to second order in the exponent (and noting that T_b is **not** a function of E for the system), following the methodology in Problem 1.3:

$$\begin{aligned} Z(T_\mathsf{b},V,N) &= \int \mathrm{d}E\ \exp\left(-\beta_\mathsf{b}[E - T_\mathsf{b}S(E,V,N)]\right) \\ &= \int \mathrm{d}E\ \exp\Bigg(- \beta_\mathsf{b}\Bigg\{[U - T_\mathsf{b}S(U,V,N)] + (E-U)\underbrace{\left[1 - T_\mathsf{b}\left(\frac{\partial S}{\partial E}\right)_V\Bigg|_{E=U}\right]}_{=0 \text{ as stationary point}} + \\ &\qquad -\frac{1}{2}(E-U)^2 T_\mathsf{b}\left(\frac{\partial^2 S}{\partial E^2}\right)_V\Bigg|_{E=U} + \cdots\Bigg\}\Bigg) \\ &\simeq \mathrm{e}^{-\beta_\mathsf{b}[[U - T_\mathsf{b}S(U,V,N)]} \int \mathrm{d}E\ \exp\left\{+\beta_\mathsf{b}\frac{1}{2}(E-U)^2 T_\mathsf{b}\left(\frac{\partial^2 S}{\partial E^2}\right)_V\Bigg|_{E=U}\right\} \ . \end{aligned}$$

The first term was a constant – there was no dependence on E, so could be removed from the integral, and we have discarded the higher order terms in the exponent.

We suspect the quadratic term is small. To see why this is indeed true, note[1] and remembering we are at constant V.

$$\frac{\partial^2 S}{\partial E^2} = \frac{\partial}{\partial E}\frac{1}{T} = -\frac{1}{T^2}\frac{\partial T}{\partial E} = -\frac{1}{T^2 C_V} \ .$$

Then the expression for Z takes the form (at this stage replacing T_b by T):

$$\begin{aligned} Z(T,V,N) &\simeq \mathrm{e}^{-\beta[U-TS(U,V,N)]} \int \mathrm{d}E \ \exp\left(-\frac{(E-U)^2}{2k_\mathrm{B}T^2C_V}\right) \\ &= \mathrm{e}^{-\beta F(T,V,N)} \int \mathrm{d}E \ \exp\left(-\frac{(E-U)^2}{2k_\mathrm{B}T^2C_V}\right) \ . \end{aligned} \tag{4.4}$$

Note that the fact that E is an extensive variable (proportional to V or N) and hence C_V is as well.

We now have a Gaussian integral, using $x = E - U$ with $\int_{-\infty}^{\infty} \mathrm{d}x \ \mathrm{e}^{-\alpha x^2} = \sqrt{\pi/\alpha}$, where $\alpha = 1/(2k_\mathrm{B}T^2C_V)$, Thus

$$\begin{aligned} Z(T,V,N) &= \mathrm{e}^{-\beta F(T,V,N)}\sqrt{2\pi k_\mathrm{B}T^2C_V} \\ \Rightarrow \quad \ln Z(T,V,N) &= -\beta F(T,V,N) + \frac{1}{2}\ln\left(2\pi k_\mathrm{B}T^2C_V\right) \ . \end{aligned}$$

Note the first term on the right hand side is proportional to N but the second term is proportional to $\ln N$, from $C_V \propto N$. In the case of $N \sim N_\mathrm{A} = 10^{23}$, Avogadro's number, $\ln N \simeq 45$, which is negligible. This is reflects the sharp peak in the integrand.

Solving for F, discarding the logarithmic term, we see

$$F(T,V,N) = -k_\mathrm{B}T \ln Z(T,V,N) \ , \tag{4.5}$$

which is a very important equation.

In problem (2.2) the following important characterisation of the small role of fluctuations is demonstrated:

$$\begin{aligned} \langle (E-U)^2 \rangle &= k_\mathrm{B}T^2C_V \propto N \\ \Rightarrow \quad \frac{\sqrt{\langle (E-U)^2\rangle}}{U} &\propto \frac{1}{\sqrt{N}} \ . \end{aligned} \tag{4.6}$$

Thus the scale of the fluctuations in energy, $\sqrt{\langle (E-U)^2\rangle} \ll U$. This shows that thermodynamics emerges from statistical mechanics as we take the so-called *thermodynamic* limit of $N \to \infty$, along with $V \to \infty$. This is reminiscent of regaining Newtonian mechanics from relativistic mechanics as $v/\mathrm{c} \to 0$, with $1/\sqrt{N}$ being the counterpart to v/c.

[1] This is the point where the distinction between T_b as an external parameter and $1/T_\mathrm{s}(E,V,N)$ for the system is important.

We may now explore this relation of the macroscopic F (encompassing all the thermodynamic information) to the microscopic world, represented by the canonical partition function Z, in some examples.

4.4 Summary

In this section we have:

- remembered the central nature of the (Helmholtz) free energy, $F(T, V, N)$;
- remembered the derivation of other thermodynamic functions from relation, such as the entropy, $S(T, V, N)$, pressure, $p(T, V, N)$ and internal energy $U(T, V, N)$ (and hence the specific heat etc);
- noted the classification of intensive (e.g. T) and extensive (e.g. N) variables;
- constructed the other thermodynamic potentials and noted when they are useful – eg Gibbs free energy, $G(T, p, N)$, for constant pressure;
- constructed the expression relating (as $N \to \infty$) the canonical partition function, $Z(T, V, N)$, and the Helmholtz free energy, $F = -k_{\mathrm{B}} T \ln Z$;
- shown the corrections are negligible as $N \to \infty$, the thermodynamic limit.

We have connected the macroscopic thermodynamic functions via F to the microscopic variables through Z.

Appendices

A Generating functions and the partition function – non-examinable

This appendix exposes the mathematical significance of the functions $W(E, V, N)$ and $Z(T, V, N)$ in statistical mechanics.

A *generating function* allows the calculation of moments – e.g. the average or variance – of a probability distribution by *differentiation*, not integration.

If we deal with (normalised) discrete probabilities, p_n, where n labels the outcome of a trial (point in the *sample space*), then we may define a generating function, $H(t)$, as:

$$H(t) = \sum_n p_n \mathrm{e}^{nt} \ .$$

Then we can see that:

$$\begin{aligned}
H(t=0) &= \sum_n p_n \times 1 && = 1 \\
H'(t) &= \sum_n n p_n \mathrm{e}^{nt} \\
\Rightarrow \quad H'(t=0) &= \sum_n n p_n && = \langle n \rangle \\
H''(t) &= \sum_n n^2 p_n \mathrm{e}^{nt} \\
\Rightarrow \quad H''(t=0) &= \sum_n n^2 p_n && = \langle n^2 \rangle \ ,
\end{aligned}$$

and so on for higher moments.

The generating function is not unique one could have used

$$\begin{aligned}
G(t) &= \sum_n p_n t^n \\
\text{Then} \quad t\frac{\mathrm{d}}{\mathrm{d}t} G(t) &= t \sum_n n p_n t^{n-1} && = \sum_n n p_n t^n \\
\Rightarrow \quad t\frac{\mathrm{d}}{\mathrm{d}t} G(t=1) &= 1 \sum_n n p_n 1^{n-1} && = \langle n \rangle \\
t\frac{\mathrm{d}}{\mathrm{d}t}\left(t\frac{\mathrm{d}}{\mathrm{d}t} G(t)\right)\bigg|_{t=1} &= 1 \sum_n n^2 p_n 1^{n-1} && = \langle n^2 \rangle \qquad .
\end{aligned}$$

One ugly aspect of both $H(t)$, and to a greater extent, $G(t)$, is that they "generate" the moments $\mu_m = \langle n^m \rangle$, but not, for example, the variance $\sigma^2 = \kappa_2 = \langle n^2 \rangle - \langle n \rangle^2$, and its counterparts for higher moments. These *cumulants* strip out the information generated by the lower moments (eg the variance is *really* the width of the distribution around the mean). They may be generated directly from $H(t)$.

Let the *cumulant generating function*, $K(t)$, be defined as

$$K(t) = \ln H(t) \ ,$$

Then we see:

$$\begin{aligned}
K'(t) &= \frac{H'(t)}{H(t)} \\
\Rightarrow \quad K'(t=0) &= \frac{H'(t=0)}{H(t=0)} && = \frac{\langle n \rangle}{1} \\
K''(t) &= \frac{H''(t)}{H(t)} - \frac{(H'(t))^2}{(H(t))^2} \\
\Rightarrow \quad K''(t=0) &= \frac{H''(t=0)}{H(t=0)} - \frac{(H'(t=0))^2}{(H(t=0))^2} && = \frac{\langle n^2 \rangle}{1} - \frac{\langle n \rangle^2}{1} \\
&= \kappa_2 \ ,
\end{aligned}$$

the variance.

The parallel with the both

$$\begin{aligned}
S(E,V,N) &= k_{\text{B}} \ln W(E,V,N) \\
\text{and} \quad F(T,V,N) &= -k_{\text{B}} T \ln Z(T,V,N) \ ,
\end{aligned}$$

is striking noting the "generation" of thermodynamic functions such as:

$$\frac{\partial S}{\partial E} = \frac{1}{T} \quad \text{and} \quad \frac{\partial F}{\partial V} = -p$$

etc.

We may regard W and Z as generating functions and S and F as "cumulant generating functions", where E and T play the roles of t in the simple cases at start of this appendix. A generalisation is that there are more parameters to differentiate – e.g. T and V (and N) for F.